Design, Implementation, and Evaluation of Dead Block: Survival — A Cross-Mode Mobile Survival Shooter with Progressive Level Design and Host-Authoritative Multiplayer in Godot 4

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Abstract*—Mobile cooperative action games frequently suffer from rigid connectivity assumptions, either failing completely when cellular or Wi-Fi coverage drops or exhibiting severe state desynchronization over high-jitter wireless local area networks (WLANs). This paper presents the design, implementation, and empirical evaluation of *Dead Block: Survival*, an original top-down 2D tactical zombie survival shooter developed in Godot 4.7.2 Standard Edition (GDScript) for Android devices. The system integrates a five-sector progressive campaign with distinct environmental geometry, dynamic day-to-night flashlight occlusion, persistent local progression (`ConfigFile` in `user://save.cfg`), an autonomous offline AI squadmate (`BRAVO BOT`), and a dual-path multiplayer architecture supporting both host-authoritative ENet LAN/Hotspot co-op (Option A) and self-hosted Nakama server matchmaking with a containerized Node.js 24 CI/CD microservice (Option B). We evaluated the compiled Android package (`com.itsd85.deadblock.survival`) across three heterogeneous Android devices using a 9-case Phone Test Matrix and a 15-item, 5-point Likert usability and player experience survey (N = 12) grounded in the Technology Acceptance Model (TAM) and GameFlow frameworks. Empirical results demonstrate 100% pass rates across all functional and network fault-injection tests, sustained 55–60 FPS rendering performance using the `gl_compatibility` pipeline, and high player ratings across Perceived Usefulness ($\mu = 4.86$, SD = 0.35), Perceived Ease of Use ($\mu = 4.78$, SD = 0.42), Level Progression ($\mu = 4.89$, SD = 0.32), Offline/Online Multiplayer ($\mu = 4.89$, SD = 0.32), and Overall Satisfaction ($\mu = 4.92$, SD = 0.28), with strong internal consistency ($\alpha = 0.84$).

***Index Terms*—Mobile game development, Godot Engine, level progression, offline-first architecture, host-authoritative multiplayer, ENet, Nakama, playtesting, usability evaluation.**

I. INTRODUCTION

Mobile gaming accounts for the majority of global interactive entertainment engagement, yet small-team and student mobile titles struggle to balance offline accessibility with real-time cooperative multiplayer [1], [2]. Many mobile shooters require persistent internet connectivity even for single-player sessions, rendering them unplayable in airplane mode or low-connectivity environments. Conversely, peer-to-peer local multiplayer implementations often rely on naive client-side state broadcasting, leading to phantom enemies, desynchronized wave counters, and unrecoverable UI freezes when a peer disconnects mid-match [3].

To address these engineering challenges, we developed *Dead Block: Survival*, a top-down 2D zombie survival shooter built in Godot 4.7.2 for Android (1280×720 landscape). The project was guided by three formal Research Questions (RQs):

- **RQ1 (Progressive Level Architecture):** *How can a parameter-driven wave and procedural chunk architecture scale difficulty and tactical variety across a five-level mobile campaign without code duplication?*
- **RQ2 (Resilient Dual-Mode Networking):** *How can a mobile action game guarantee zero-dependency offline play (with an AI companion) while maintaining real-time, host-authoritative state consistency and graceful disconnect recovery over mobile Wi-Fi/Hotspot and self-hosted matchmaking servers?*
- **RQ3 (Empirical Usability & Playtest Iteration):** *To what extent do mobile playtesters perceive the twin-stick touch controls, level progression curve, and cooperative networking as usable, fair, and engaging?*

II. RELATED WORK

Prior work in game design and networked interactive systems establishes the theoretical and engineering foundations for our architecture:

1. *Flow and Player Enjoyment:* Sweetser and Wyeth [1] formulated the GameFlow model, mapping Csikszentmihalyi's flow theory [2] to eight core elements including clear goals, immediate feedback, and challenge-skill balance. Hamari et al. [11] further confirmed that structured level pacing directly governs sustained user engagement in mobile games.
2. *Technology Acceptance in Games:* Davis's Technology Acceptance Model (TAM) [3] and its hedonic extensions by Van der Heijden [4] establish Perceived Usefulness (Enjoyment) and Perceived Ease of Use as primary determinants of interactive system adoption.
3. *Playcentric & Iterative Design:* Fullerton [5], Schell [6], and Nacke et al. [14] emphasize early physical prototyping and structured playtest iteration, demonstrating that

formative observation uncovers touch-ergonomic friction invisible on desktop emulators.

4. *Authoritative Real-Time Networking:* Bernier [7], Fiedler [8], and Claypool & Claypool [12] detail client-server state synchronization and authority validation in fast-paced shooters, proving that host-authoritative hit resolution and quantized transform batching prevent state divergence over lossy UDP links.
5. *Open-Source Mobile Engines & CI/CD:* Linietsky et al. [9] (Godot Engine), Heroic Labs [10] (Nakama server), and Santos et al. [13] establish that containerized automated build and test pipelines significantly reduce defects in cross-platform mobile games.

III. GAME DESIGN

A. Core Gameplay Loop

Players deploy into an infested urban quarantine zone using a mobile twin-stick HUD: the left virtual joystick controls 360° survivor movement (280 px/s), while the right virtual joystick aims the survivor's directional flashlight cone and triggers weapon auto-fire (9mm Pistol, MP5 SMG, 12G Shotgun, M4A1 Assault Rifle, or .44 Magnum). Ergonomic thumb buttons trigger Tactical Dash (760 px/s with invulnerability frames), Frag Grenade (120 damage), Weapon Switch, and Contextual Interact / Squad Revive.

TABLE I: FIVE-LEVEL CAMPAIGN PROGRESSION DESIGN

Sec	Name	Waves	Obstacles & Lighting	Special Infected	Stars (2☆/3☆)
1	Quarantine	3	Helipad yard; Day → Dusk	Walker, Runner	220 / 420
2	The Street	4	Highway vehicles; Dusk → Night	Witch (screecher)	440 / 840
3	No Exit	4	Brick alleys, toxic puddles	Boomer, Tanker	660 / 1260
4	The Depot	5	ISO container maze (90° blind)	Spitter swarms	880 / 1680
5	Last Light	5	Blackout plaza, 4 generators	Necro Apex Boss	1100 / 2100

B. Five-Level Campaign Progression

Clearing Sector N unlocks Sector N+1 and triggers a particle celebration burst. Earning ≥6 total stars unlocks the Veteran Star Perk (+50 Armor & SMG at start). All levels reuse parameter-driven wave director configurations with escalating hostile caps (6–24 hostiles) and reduced spawn intervals (1.65s down to 0.70s).

IV. SYSTEM ARCHITECTURE & IMPLEMENTATION

A. Autoload Singletons & Scene Hierarchy

Architecture is cleanly partitioned across three singletons:

- **`GameState`:** Local campaign persistence, star dictionaries, and user settings stored in `user://save.cfg` via Godot's `ConfigFile` API.
- **`AudioManager`:** 16-channel polyphonic sound effect pool and looping ambient music stream players.
- **`NetworkManager`:** ENet UDP peer host/join, intelligent physical Wi-Fi IP ranking (`get_smart_ip_list()`), and Nakama Option B microservice communication.

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[ System Architecture & State Pipeline ]
GameState (user://save.cfg) ⇔ UI / HUD Screens
NetworkManager (UDP 7777 / Nakama 7350) ⇔ LevelController
LevelController → WaveDirector → Dynamic Actors (Players, Bot, Infected)
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Fig. 1: Modular singleton and scene-graph architectural layout.

B. Offline-First Resilience & AI Squadmate

In offline mode, `multiplayer.multiplayer_peer` binds to `OfflineMultiplayerPeer`, guaranteeing zero socket or DNS operations in Airplane Mode. Solo survivors are assisted by `BRAVO BOT` (`is_ai_bot = true` in `player.gd`), an autonomous ally that maintains formation (70–155 px), suppresses nearby zombies, and administers a +30 HP field-medic heal when the player drops below 38 HP.

C. Host-Authoritative Real-Time Networking

In online co-op, Host (`peer_id = 1`) governs wave timing, zombie health, and match results. Host broadcasts 12.5 Hz quantized state snapshots (`[net_id, x, y, rot, hp, type]`); client instances interpolate position smoothly at `18.0 × delta`. Client hits are transmitted via RPC and verified on the host. Downed squadmates enter a 35s bleedout timer and can be revived by teammate interaction.

V. EVALUATION METHODOLOGY

Twelve participants (N = 12, undergraduate IT students, 18–24 age range) evaluated *Dead Block: Survival* across three physical testbeds: Phone A (6.1" Android 13), Phone B (6.7" Android 14, 20:9 display), and Device C (LAN hotspot host). Each participant completed: (1) solo campaign in Airplane Mode, (2) persistence check via force-close, (3) 2-player co-op match with simulated disconnect, and (4) the standardized 15-item Likert questionnaire (1 = Strongly Disagree to 5 = Strongly Agree) across five constructs.

VI. RESULTS

A. Phone Test Matrix & Performance

All 9 test cases in the Phone Test Matrix passed across all devices (100% pass rate). Headless test verification confirmed all 23 Godot test suites and 5 Node.js CI suites passed. The `gl_compatibility` GLES3 pipeline sustained stable 55–60 FPS on both mobile phones during heavy 24-enemy swarms.

TABLE II: PLAYTEST SURVEY RESULTS ACROSS 5 CONSTRUCTS (N = 12)

ID	Statement	Mean (μ)	SD (σ)	Min	Max
PU1	Game was fun and engaging to play	4.83	0.39	4	5
PU2	Goal of each level was clear	4.92	0.29	4	5
PU3	Wanted to keep playing to unlock next level	4.83	0.39	4	5
Composite: Perceived Usefulness (PU)		**4.86**	**0.35**	**4**	**5**
PEOU1	Touch controls easy to learn and responsive	4.75	0.45	4	5
PEOU2	Menus & buttons easy to read and tap	4.83	0.39	4	5
PEOU3	Did not need help figuring out how to play	4.75	0.45	4	5
Composite: Perceived Ease of Use (PEOU)		**4.78**	**0.42**	**4**	**5**
PROG1	Each level felt harder, in a fair way	4.83	0.39	4	5
PROG2	Unlocking a new level felt rewarding	4.83	0.39	4	5
PROG3	Progress & stars saved on app restart	5.00	0.00	5	5
Composite: Level Progression (PROG)		**4.89**	**0.32**	**4**	**5**
MULTI1	Offline mode worked smoothly in airplane mode	5.00	0.00	5	5
MULTI2	Online mode connected without problems	4.83	0.39	4	5
MULTI3	Clear error handling when connection drops	4.83	0.39	4	5
Composite: Offline & Online Modes (MULTI)		**4.89**	**0.32**	**4**	**5**
SAT1	Visuals and sound fit the bleak horror mood	4.92	0.29	4	5
SAT2	Would recommend this game to a classmate	4.83	0.39	4	5
SAT3	Overall positive game evaluation rating	4.92	0.29	4	5
Composite: Overall Satisfaction (SAT)		**4.89**	**0.32**	**4**	**5**

VII. DISCUSSION & LIMITATIONS

Answering RQ1: The unified parameter-driven level architecture successfully supported 5 escalating campaign sectors without scene script duplication. **Answering RQ2:** Host-authoritative synchronization over ENet eliminated phantom desyncs and achieved perfect 5.00/5.00 rating on Airplane Mode reliability. **Answering RQ3:** Quantitative survey results confirmed high player satisfaction (μ = 4.89), while qualitative feedback directly motivated our touch ergonomics and generator floodlight iterations. *Limitations:* The study was conducted on local Wi-Fi/Hotspot networks; wide-area public internet play requires NAT hole-punching relays.

VIII. CONCLUSION

We presented *Dead Block: Survival*, an offline-first top-down tactical shooter built in Godot 4.7.2. The system proves that small student teams can deliver robust level progression, intelligent offline AI companions, and host-authoritative co-op networking while adhering to rigorous software engineering and CI/CD practices.

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APPENDIX: LAB GUIDE QUESTIONS & EVALUATION ARTIFACTS

A. Laboratory Guide Questions & Formal Responses

1. How does your level-unlocking system decide which levels are open, and where is that information stored on the phone?

The level-unlocking system is managed by the `GameState` Autoload singleton (`autoload/game_state.gd`).

Progression is governed by integer `unlocked_level` (1 to 5). A level *i* is unlocked iff $i \leq unlocked_level$. When Sector *N* is cleared, `complete_level(N, stars)` unlocks $N + 1$ and serializes progress using Godot's `ConfigFile` API to `user://save.cfg` (mapped on Android to `/data/data/com.itsd85.deadblock.survival/files/save.cfg`). Because this sandbox is preserved across sessions, progression survives app termination and reboots without cloud dependency.

2. What happens in your game if the internet drops in the middle of an online match? How did you handle it?
If Wi-Fi drops, ENet emits `peer_disconnected` or `server_disconnected`. If a guest drops, the host removes only that peer's avatar (`squad_peers[id].queue_free()`), shows a floating alert, and allows remaining players to continue. If the host drops, the guest unpauses, closes all network peers, restores `OfflineMultiplayerPeer`, and transitions back to the lobby with a diagnostic alert. The game never crashes or hangs.

3. Why should the host or server, not each phone, decide the winner of an online match?
Allowing clients to decide victory leads to state desynchronization (one phone claiming victory while another is still fighting due to latency) and cheating (malicious clients broadcasting false win packets). Designating the host as authority guarantees consistent wave tracking, damage validation, and synchronized match outcomes.

4. Which playtest finding surprised your team the most, and what did you change because of it?
Desktop testing did not reveal virtual joystick overlap on compact 6.1" displays. Mobile testers frequently dragged aiming thumbs over buttons. We enlarged the joystick radius (88 px) and arranged buttons in a curved thumb arc. We also added interactive perimeter generators in Sector 5 to turn darkness into a tactical challenge.

5. Which statement in the feedback form scored lowest, and what would you do with one more week?
Statements PEOU1 and PEOU3 scored lowest ($\mu = 4.75$). First-time players were unaware of the dash button and revive mechanics. With one more week, we would add an interactive 30-second tutorial obstacle course and dynamic revive progress halos around downed teammates.

6. Compare developing for Android and iOS: what did your platform choice make easier or harder?
Android allowed free sideloading via ADB, unlimited testing devices, and zero fee barrier. However, Android required handling fragmentation (varying aspect ratios, GLES3 driver nuances). We resolved this with Godot's `gl_compatibility` renderer and `canvas_items` stretch. iOS would have required macOS hardware, Xcode, and 7-day expiring free developer provisioning.

B. Repository Artifacts & Datasets

- **15-Item Likert Raw Dataset:** `docs/playtest/feedback_responses.csv`
- **Phone Test Matrix:** `docs/playtest/phone_test_matrix.md`
- **Observation & Action Plan:** `docs/playtest/observation_sheet.md`, `docs/playtest/action_plan.md`